

Transformation of Energy from One Form to Another

Aim:-

To demonstrate the conversion of potential energy to the total kinetic energy (Translational KE + Rotational KE) for a rolling cylinder.

Apparatus:

- Smartphone with App Phyphox,
- Cylinder (90mm pvc pipe/chip can) and scrap sponge for padding
- Inclined plane arrangement, scale.

Principle:

The total energy before rolling = potential energy = Mgh .



Total energy as the cylinder reaches at the bottom of the inclined plane,

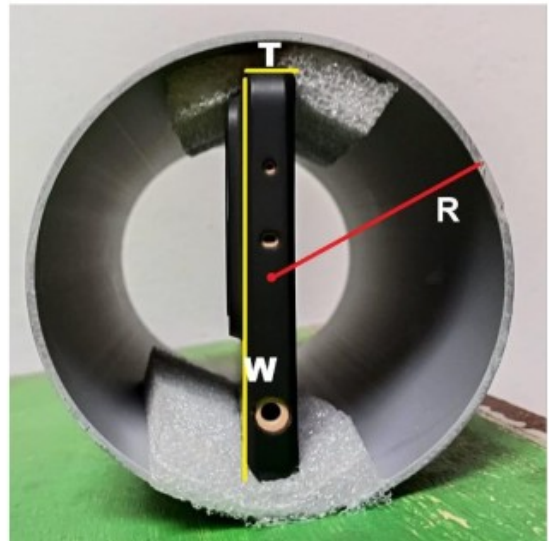
$$KE_{Total} = KE_{Translational} + KE_{Rotational}$$

$$KE_{Total} = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$$

$$M = M_{Cylinder} + M_{Phone}$$

$$I = I_{Cylinder} + I_{Phone}$$

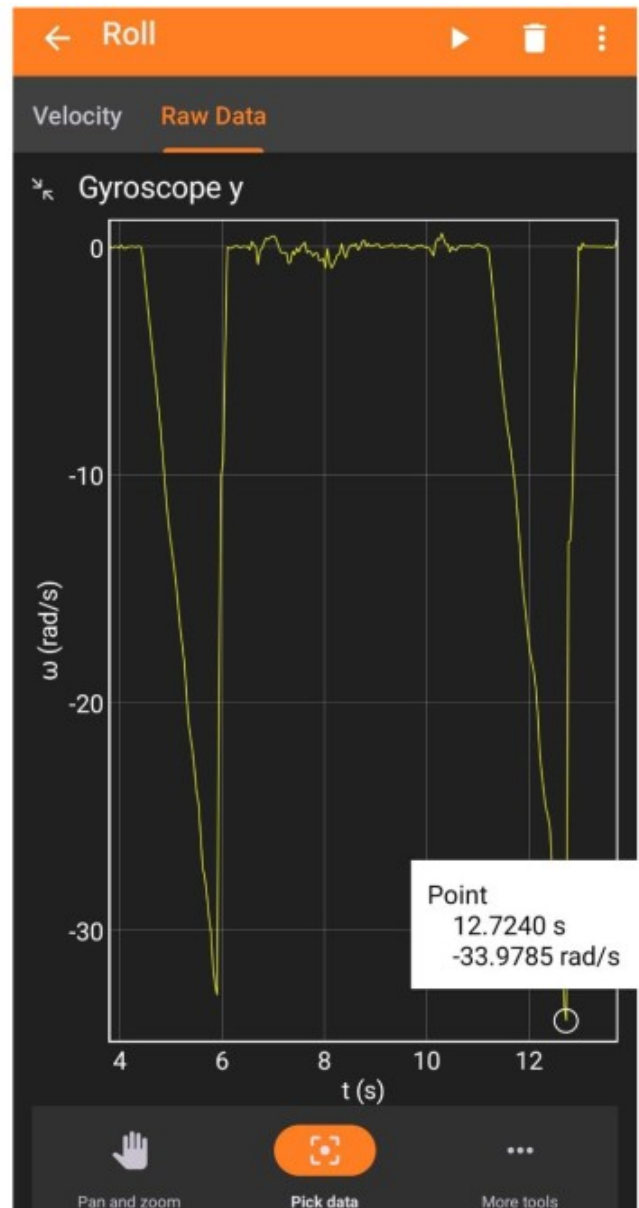
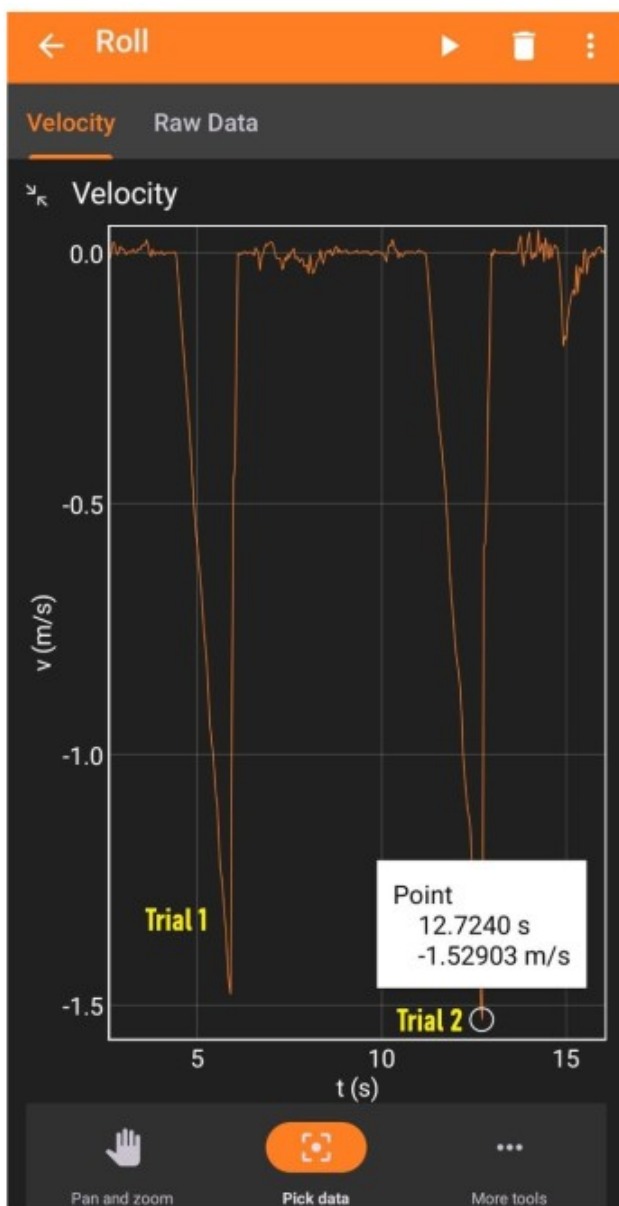
$$I = \frac{1}{2}M_c R^2 + \frac{M_p(T^2 + W^2)}{12}$$



Procedure:

- Setup an inclined plane with approximately 1m length.
- Open Phyphox > Mechanics > Roll.
- Measure the outer radius of the cylinder and give the value of Roll radius (in cm).

- Set the phone inside the cylinder with suitable padding using paper or sponge, so that the phone is symmetrically aligned along the axis of the cylinder.
- Place the cylinder at the top of the inclined plane and measure the height, h (upto the line of contact of the roll and plane)
- Tap the play button and release the cylinder from the height, without pushing. A cushion like arrangement can be placed at the bottom of the plane to stop the roll. (The remote access option may be used control the app from a second device-phone/laptop)
- Repeat the experiment (this can be done in the recoding mode along with the data of first trial).
- Tap the velocity(m/s) graph and suitable scale using the fingers. The velocity increases from zero(roll is at top) and reaches a maximum(at bottom).
- Use 'Pick data' tool to get the maximum value of v .
To get the value of ω , move to 'Raw Data' tab. Take the average of the two trials.
(The value of ω , and hence v , can be positive or negative depending on whether the phone is rolling clockwise or anticlockwise)



Observations:

Outer radius of the cylinder, $R = 4.5\text{cm}$

Width of the phone, $w = 7.5\text{cm}$

Thickness of the phone, $T = 0.75\text{cm}$

Mass of the phone, $M_{\text{Phone}} = 205\text{gm}$

Mass of the cylinder $M_{\text{Cylinder}} = 150\text{gm}$

$M = M_{\text{Cylinder}} + M_{\text{Phone}} = 0.335\text{Kg}$

Results:

Demonstrated and verified the conversion of potential energy to the total kinetic energy of the rolling cylinder.